Predicting Student Dropout with Gradient Boosting

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Methods

- We trained XGBoost on 48,312 student records from 2015-2024 using 112 engir
- Hyperparameters were tuned with 5-fold nested cross-validation and Bayesian
- Class imbalance (8.4% dropout) was handled with focal loss and stratified sam
- Calibration: isotonic regression on a held-out 2024 cohort
- Fairness audit across gender, nationality and first-generation status
- SHAP values for global and per-student explanations
- Sensitivity analysis excluding the COVID-19 terms 2020-2021

Results

Model 1 AUC 0.70 Brier 0.19 ECE 0.00 n=48312 folds=5 seed=0

Model 2 AUC 0.71 Brier 0.18 ECE 0.01 n=48312 folds=5 seed=7

Model 3 AUC 0.72 Brier 0.17 ECE 0.02 n=48312 folds=5 seed=14

Model 4 AUC 0.73 Brier 0.16 ECE 0.03 n=48312 folds=5 seed=21

Model 5 AUC 0.74 Brier 0.15 ECE 0.04 n=48312 folds=5 seed=28

Model 6 AUC 0.75 Brier 0.14 ECE 0.05 n=48312 folds=5 seed=35

Model 7 AUC 0.76 Brier 0.13 ECE 0.06 n=48312 folds=5 seed=42

Model 8 AUC 0.77 Brier 0.12 ECE 0.07 n=48312 folds=5 seed=49

Model 9 AUC 0.78 Brier 0.11 ECE 0.08 n=48312 folds=5 seed=56

Model 10 AUC 0.79 Brier 0.19 ECE 0.00 n=48312 folds=5 seed=63

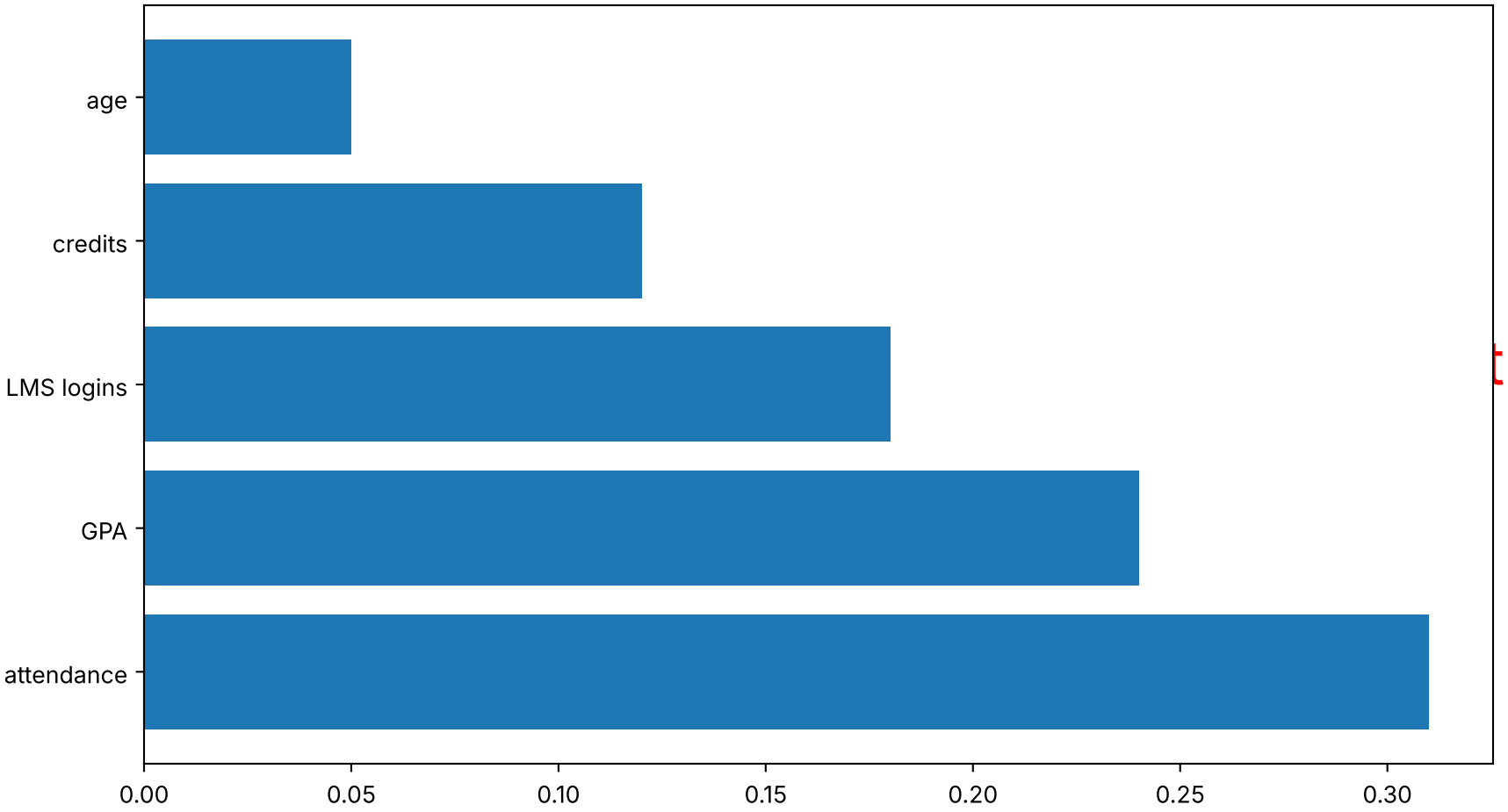
Model 11 AUC 0.80 Brier 0.18 ECE 0.01 n=48312 folds=5 seed=70

Model 12 AUC 0.81 Brier 0.17 ECE 0.02 n=48312 folds=5 seed=77

Model 13 AUC 0.82 Brier 0.16 ECE 0.03 n=48312 folds=5 seed=84

Model 14 AUC 0.83 Brier 0.15 ECE 0.04 n=48312 folds=5 seed=91

Feature importance



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